HOW IT WORKS · SUPPLY, DEMAND, ECOSYSTEM

Reclaimed Lumber Intelligence Layer

Open the live app ›

How the app turns open data and sourced coefficients into a forecast of salvageable lumber, then maps the demand and the ecosystem that decide whether it gets reused.

The pipeline, end to end

Six stages. Each box feeds the next, and every coefficient carries a citation and a range.

01
Sourcing & inputs
- Toronto Open Data (live permits)
- StatCan Census (dwellings, age)
- StatCan Building Permits
- Coefficients (FPL, Oregon DEQ, MPAC)

02
Canonical table
- One schema for every city
- Void and coverage flags
- Real versus modelled

03
Recovery cascade
- Framing lumber in the building
- Recoverable, then salvageable
- Spec-ready, then value
- One sourced coefficient per step

04
Uncertainty
- Monte Carlo over every range
- P10 / P50 / P90 bands
- Tornado: the lever that matters

05
Outputs
- Supply and demand forecasts to 2036
- Maps, three-gap analysis, embodied carbon
- Value, scenarios, confidence scores

06
Demand & ecosystem
- Demand drivers, segments and economics
- Two-sided ecosystem: firms and buyers
- Registries that feed matchmaking

Inside the recovery cascade

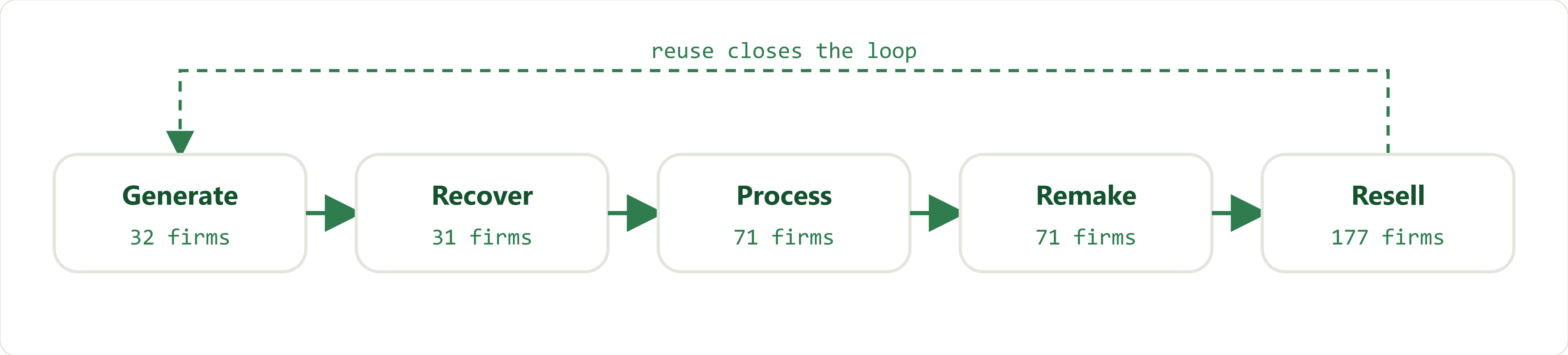
One post-war single-detached home: 120 m2 of floor area, built around 1965, mixed demolition practice. Each arrow multiplies by a sourced coefficient.

Floor area 120 m2 › **Framing** 9,290 bf 79 bf/m2 › **Recoverable** 2,029 bf x0.30 x0.73 › **Salvageable** 1,420 bf x0.70 › **Spec-ready** 781 bf x0.55 › **Value** $3,750 $4.80/bf

Coefficients: framing content (McKeever & Phelps, FPL 1994); recovery and condition (Oregon DEQ 2019); denailing and grading (Falk, FPL-RP-650, 2008).

Two-sided ecosystem and demand

Supply is half the story. This side covers the firms that recover and remake wood, and the buyers and channels that absorb it. The value chain runs from generation to resale and loops back, and the demand side is now logged alongside supply.



Firm counts are the national SME census by value-chain step (Light House for ECCC, March 2026; 252 SMEs). Retail leads because it includes 102 Habitat ReStores.

NAMED COMPANIES
58

NATIONAL SMES
252

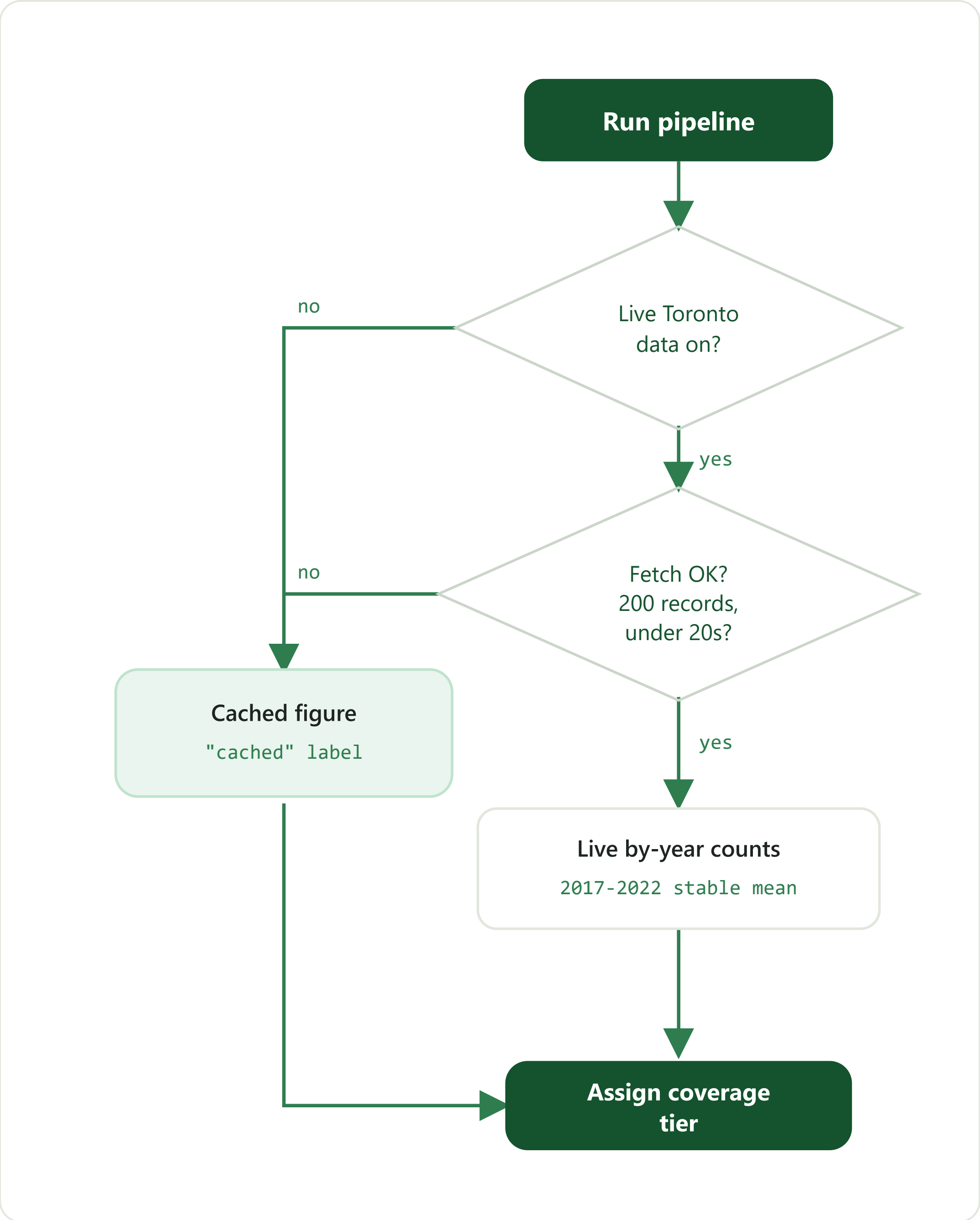
DEMAND LEGAL TODAY
~22M bf

DEMAND IF REUSE ALLOWED
~88M bf

Latent demand runs above current spec-ready supply, so the binding constraint is supply and coordination. The diagnosis below ranks why.

#	Why recoverable wood is not reclaimed	Binds on
1	Structural-reuse code wall (salvaged lumber must be regraded; no code path)	Demand (rules)
2	No forward supply signal, so buyers cannot commit early	Coordination
3	Processing and de-nailing labour cost	Supply
4	Deconstruction costs more and takes longer than demolition	Supply
5	Storage, logistics, and missing grading standards	Supply

Decision: live data or fallback



Coverage sets the band

Coverage	Markets	Band
HIGH	Toronto (live feed)	+/- 10%
MEDIUM	Vancouver, Montreal, Calgary, Edmonton, Ottawa-Gatineau	+/- 25%
LOW	The other 19 CMAs (inferred)	+/- 45%

What comes out

Output	Where
Supply and demand forecasts, P10/P50/P90	Overview, Municipal, Forecast, Demand drivers
Sensitivity tornado	Forecast & uncertainty
Two-sided ecosystem and three-gap analysis	Ecosystem, Supply & demand gaps
Demand drivers, segments and economics	Demand drivers, Economics
Embodied carbon and policy	Policy & carbon
Supply and demand registries, matchmaking	Platform